

Q1 - 25 July - Shift 1

Let ABC be a triangle such that $\overrightarrow{BC} = \vec{a}$, $\overrightarrow{CA} = \vec{b}$,

$\overrightarrow{AB} = \vec{c}$, $|\vec{a}| = 6\sqrt{2}$, $|\vec{b}| = 2\sqrt{3}$ and $\vec{b} \cdot \vec{c} = 12$

Consider the statements :

(S1) : $|\vec{a} \times \vec{b}| + |\vec{c} \times \vec{b}| - |\vec{c}| = 6(2\sqrt{2} - 1)$

(S2) : $\angle ABC = \cos^{-1}\left(\sqrt{\frac{2}{3}}\right)$. Then

(A) both (S1) and (S2) are true

(B) only (S1) is true

(C) only (S2) is true

(D) both (S1) and (S2) are false

Space for your notes:

Q2 - 25 July - Shift 2

Let $\vec{a} = \hat{i} - \hat{j} + 2\hat{k}$ and $\vec{b}$ be a vector such that

$\vec{a} \times \vec{b} = 2\hat{i} - \hat{k}$ and $\vec{a} \cdot \vec{b} = 3$. Then the projection

of $\vec{b}$ on the vector $\vec{a} - \vec{b}$ is :-

(A) $\frac{2}{\sqrt{21}}$ (B) $2\sqrt{\frac{3}{7}}$

(C) $\frac{2}{3}\sqrt{\frac{7}{3}}$ (D) $\frac{2}{3}$

Space for your notes:

Q3 - 26 July - Shift 1

Let $\vec{a} = \alpha\hat{i} + \hat{j} - \hat{k}$ and $\vec{b} = 2\hat{i} + \hat{j} - \alpha\hat{k}$, $\alpha > 0$. If

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the projection of $\vec{a} \times \vec{b}$ on the vector $-\hat{i} + 2\hat{j} - 2\hat{k}$

is 30, then α is equal to

(A) $\frac{15}{2}$

(B) 8

(C) $\frac{13}{2}$

(D) 7

Q4 - 27 July - Shift 1

Let $\vec{a} = \alpha\hat{i} + \hat{j} + \beta\hat{k}$ and $\vec{b} = 3\hat{i} - 5\hat{j} + 4\hat{k}$ be two

Space for your notes:

vectors, such that $\vec{a} \times \vec{b} = -\hat{i} + 9\hat{j} + 12\hat{k}$. Then the

projection of $\vec{b} - 2\vec{a}$ on $\vec{b} + \vec{a}$ is equal to

(A) 2

(B) $\frac{39}{5}$

(C) 9

(D) $\frac{46}{5}$

Q5 - 27 July - Shift 1

Let $\vec{a} = 2\hat{i} - \hat{j} + 5\hat{k}$ and $\vec{b} = \alpha\hat{i} + \beta\hat{j} + 2\hat{k}$. If

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$\left((\vec{a} \times \vec{b}) \times \hat{i}\right) \cdot \hat{k} = \frac{23}{2}$, then $|\vec{b} \times 2\hat{j}|$ is equal to

(A) 4

(B) 5

(C) $\sqrt{21}$

(D) $\sqrt{17}$

Q6 - 27 July - Shift 2

Questions

MathonGo

Let $\vec{a}$, $\vec{b}$, $\vec{c}$ be three non-coplanar vectors such that

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$$\vec{a} \times \vec{b} = 4\vec{c}, \vec{b} \times \vec{c} = 9\vec{a} \text{ and } \vec{c} \times \vec{a} = \alpha\vec{b}, \alpha > 0.$$

If $\frac{|\vec{a}| + |\vec{b}| + |\vec{c}|}{36} = \frac{1}{36}$, then α is equal to _____.

Q7 - 28 July - Shift 1

Let the vectors $\vec{a} = (1+t)\hat{i} + (1-t)\hat{j} + \hat{k}$,

Space for your notes:

$\vec{b} = (1-t)\hat{i} + (1+t)\hat{j} + 2\hat{k}$ and $\vec{c} = t\hat{i} - t\hat{j} + \hat{k}$, $t \in \mathbb{R}$

be such that for $\alpha, \beta, \gamma \in \mathbb{R}$, $\alpha\vec{a} + \beta\vec{b} + \gamma\vec{c} = \vec{0}$

$\Rightarrow \alpha = \beta = \gamma = 0$. Then, the set of all values of t is :

(A) a non-empty finite set

(B) equal to $\mathbb{N}$

(C) equal to $\mathbb{R} - \{0\}$

(D) equal to $\mathbb{R}$

Q8 - 28 July - Shift 1

Let a vector $\vec{a}$ has a magnitude 9. Let a vector $\vec{b}$ be such that for every $(x, y) \in \mathbb{R} \times \mathbb{R} - \{(0, 0)\}$, the

Space for your notes:

vector $(x\vec{a} + y\vec{b})$ is perpendicular to the vector $(6y\vec{a} - 18x\vec{b})$. Then the value of $|\vec{a} \times \vec{b}|$ is equal to:

(A) $9\sqrt{3}$ (B) $27\sqrt{3}$ (C) 9 (D) 81

Q9 - 28 July - Shift 2

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Questions

MathonGo

Let S be the set of all $a \in \mathbb{R}$ for which the angle between the vectors $\vec{u} = a(\log_e b)\hat{i} - 6\hat{j} + 3\hat{k}$ and $\vec{v} = (\log_e b)\hat{i} + 2\hat{j} + 2a(\log_e b)\hat{k}$, ($b > 1$) is acute.

Then S is equal to

- (A) $\left(-\infty, -\frac{4}{3}\right)$ (B) Φ
 (C) $\left(-\frac{4}{3}, 0\right)$ (D) $\left(\frac{12}{7}, \infty\right)$

Space for your notes:

Q10 - 29 July - Shift 1

Let $\vec{a} = 3\hat{i} + \hat{j}$ and $\vec{b} = \hat{i} + 2\hat{j} + \hat{k}$. Let $\vec{c}$ be a vector satisfying $\vec{a} \times (\vec{b} \times \vec{c}) = \vec{b} + \lambda \vec{c}$. If

$\vec{b}$ and $\vec{c}$ are non-parallel, then the value of λ is :

- (A) -5 (B) 5
 (C) 1 (D) -1

Space for your notes:

Q11 - 29 July - Shift 1

Let $\hat{a}$ and $\hat{b}$ be two unit vectors such that the angle between them is $\frac{\pi}{4}$. If θ is the angle between

the vectors $(\hat{a} + \hat{b})$ and $(\hat{a} + 2\hat{b} + 2(\hat{a} \times \hat{b}))$,

then the value of $164 \cos^2 \theta$ is equal to :

- (A) $90 + 27\sqrt{2}$ (B) $45 + 18\sqrt{2}$
 (C) $90 + 3\sqrt{2}$ (D) $54 + 90\sqrt{2}$

Space for your notes:

Q12 - 29 July - Shift 2

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Questions

MathonGo

If $(2, 3, 9)$, $(5, 2, 1)$, $(1, \lambda, 8)$ and $(\lambda, 2, 3)$ are coplanar, then the product of all possible values of λ is:

(A) $\frac{21}{2}$

(B) $\frac{59}{8}$

(C) $\frac{57}{8}$

(D) $\frac{95}{8}$

Space for your notes:

Q13 - 29 July - Shift 2

Let $\vec{a}, \vec{b}, \vec{c}$ be three coplanar concurrent vectors such that angles between any two of them is same.

If the product of their magnitudes is 14 and

$$(\vec{a} \times \vec{b}) \cdot (\vec{b} \times \vec{c}) + (\vec{b} \times \vec{c}) \cdot (\vec{c} \times \vec{a}) + (\vec{c} \times \vec{a}) \cdot (\vec{a} \times \vec{b}) = 168$$

then $|\vec{a}| + |\vec{b}| + |\vec{c}|$ is equal to:

(A) 10

(B) 14

(C) 16

(D) 18

Space for your notes:

Q14 - 29 July - Shift 2

Let $\vec{a}$ and $\vec{b}$ be two vectors such that

$$|\vec{a} + \vec{b}|^2 = |\vec{a}|^2 + 2|\vec{b}|^2, \vec{a} \cdot \vec{b} = 3 \quad \text{and} \quad |\vec{a} \times \vec{b}|^2 = 75.$$

Then $|\vec{a}|^2$ is equal to _____.

Space for your notes:

Answer Key

Q1 (C)

Q2 (A)

Q3 (D)

Q4 (D)

Q5 (B)

Q6 (36)

Q7 (C)

Q8 (B)

Q9 (B)

Q10 (A)

Q11 (A)

Q12 (D)

Q13 (C)

Q14 (14)

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Q1 (C)

$$\vec{a} + \vec{b} + \vec{c} = \vec{0}$$

$$\vec{b} + \vec{c} = -\vec{a}$$

$$|\vec{b}|^2 + |\vec{c}|^2 + 2\vec{b} \cdot \vec{c} = |\vec{a}|^2$$

$$|\vec{c}|^2 = 36$$

$$|\vec{c}| = 6$$

$$S1 : |\vec{a} \times \vec{b} + \vec{c} \times \vec{b}| - |\vec{c}|$$

$$|(\vec{a} + \vec{c}) \times \vec{b}| - |\vec{c}|$$

$$|-\vec{b} \times \vec{b}| - |\vec{c}|$$

$$0 - 6 = -6$$

$$S2 : \vec{a} + \vec{b} + \vec{c} = \vec{0}$$

$$\vec{b} + \vec{c} = -\vec{a}$$

$$|\vec{a}|^2 + |\vec{b}|^2 - 2|\vec{a}||\vec{b}|\cos(\angle ACB) = |\vec{c}|^2$$

$$\cos(\angle ACB) = \sqrt{\frac{2}{3}}$$

Q2 (A)

(A)

Q3 (D)

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$$\vec{a} \times \vec{b} = (1 - \alpha)\hat{i} + (\alpha^2 - 2)\hat{j} + (\alpha - 2)\hat{k}$$

Projection of $\vec{a} \times \vec{b}$ on $-\hat{i} + 2\hat{j} - 2\hat{k}$

$$= \frac{(\vec{a} \times \vec{b}) \cdot (-\hat{i} + 2\hat{j} - 2\hat{k})}{\sqrt{1 + 4 + 4}} = 30$$

$$\Rightarrow 2\alpha^2 - \alpha - 91 = 0$$

$$\Rightarrow \alpha = 7, -\frac{13}{2}$$

Q4 (D)

$$\text{Let } \vec{a} = \alpha \hat{i} + \hat{j} + \beta \hat{k}, \vec{b} = 3\hat{i} - 5\hat{j} + 4\hat{k}$$

$$\vec{a} \times \vec{b} = -\hat{i} + 9\hat{j} + 12\hat{k}$$

$$\Rightarrow \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \alpha & 1 & \beta \\ 3 & -5 & 4 \end{vmatrix}$$

$$\Rightarrow (4 + 5\beta)\hat{i} + (3\beta - 4\alpha)\hat{j} + (-5\alpha - 3)\hat{k}$$

$$= -\hat{i} + 9\hat{j} + 12\hat{k}$$

$$\therefore 4 + 5\beta = -1, 3\beta - 4\alpha = 9, -5\alpha - 3 = 12$$

$$\beta = -1, \alpha = -3$$

$$\therefore \vec{a} = -3\hat{i} + \hat{j} - \hat{k}, \vec{b} = 3\hat{i} - 5\hat{j} + 4\hat{k}$$

$$\therefore \vec{a} + \vec{b} = -4\hat{j} + 3\hat{k}$$

$$|\vec{a}|^2 = 11, |\vec{b}|^2 = 50$$

$$\vec{a} \cdot \vec{b} = -9 + (-5) - 4 = -18$$

$$\therefore \text{Projection of } (\vec{b} - 2\vec{a}) \text{ on } \vec{a} + \vec{b} \text{ is}$$

$$\frac{(\vec{b} - 2\vec{a}) \cdot (\vec{a} + \vec{b})}{|\vec{a} + \vec{b}|}$$

$$= \frac{|\vec{b}|^2 - 2|\vec{a}|^2 - (\vec{a} \cdot \vec{b})}{|\vec{a} + \vec{b}|} = \frac{50 - 22 - (-18)}{5} = \frac{46}{5}$$

$$\text{Ans. } \left(\frac{46}{5} \right)$$

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Q5 (B)

$$\vec{a} = 2\hat{i} - \hat{j} + 5\hat{k}, \quad \vec{b} = \alpha\hat{i} + \beta\hat{j} + 2\hat{k}$$

$$\left((\vec{a} \times \vec{b}) \times \hat{i} \right) \cdot \hat{k} = \frac{23}{2}, \text{ then } |\vec{b} \times 2\hat{j}| \text{ is}$$

$$\left((\vec{a} \cdot \hat{i})\vec{b} - (\vec{b} \cdot \hat{i})\vec{a} \right) \cdot \hat{k} = \frac{23}{2}$$

$$(\vec{a} \cdot \hat{i})(\vec{b} \cdot \hat{i}) - (\vec{b} \cdot \hat{i})(\vec{a} \cdot \hat{k}) = \frac{23}{2}$$

$$2 \times 2 - \alpha \times 5 = \frac{23}{2} \Rightarrow 5\alpha = 4 - \frac{23}{2} \Rightarrow \alpha = -\frac{3}{2}$$

$$\vec{b} \times 2\hat{j} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \alpha & \beta & 2 \\ 0 & 2 & 0 \end{vmatrix} = -4\hat{i} + 2\alpha\hat{k}$$

$$\therefore |\vec{b} \times 2\hat{j}| = \sqrt{16 + 4\alpha^2} = \sqrt{16 + 4 \times \frac{9}{4}} = 5$$

Q6 (36)

$$\vec{a} \times \vec{b} = 4\vec{c} \Rightarrow \vec{a} \cdot \vec{c} = 0 = \vec{b} \cdot \vec{c}$$

$$\vec{b} \times \vec{c} = 9\vec{a} \Rightarrow \vec{a} \cdot \vec{b} = 0 = \vec{a} \cdot \vec{c}$$

$\therefore \vec{a}, \vec{b}, \vec{c}$ are mutually $\perp$ set of vectors.

$$\Rightarrow |\vec{a}||\vec{b}| = 4|\vec{c}|, |\vec{b}||\vec{c}| = 9|\vec{a}| \text{ \& } |\vec{c}||\vec{a}| = \alpha|\vec{b}|$$

$$\Rightarrow \frac{|\vec{a}|}{|\vec{c}|} = \frac{4|\vec{c}|}{9|\vec{a}|}$$

$$\Rightarrow \frac{|\vec{c}|}{|\vec{a}|} = \frac{3}{2}$$

$$\therefore \text{If } |\vec{a}| = \lambda, |\vec{c}| = \frac{3\lambda}{2} \text{ \& } |\vec{b}| = 6$$

$$\text{Now } |\vec{a}| + |\vec{b}| + |\vec{c}| = \frac{1}{36}$$

$$\Rightarrow \frac{5}{2}\lambda + 6 = \frac{1}{36}, \lambda = \frac{-43}{18} = |\vec{a}|$$

which gives negative value of λ or $|\vec{a}|$ which is NOT possible & hence data seems to be wrong.

$$\text{But if } |\vec{a}| + |\vec{b}| + |\vec{c}| = 36$$

$$\frac{5}{2}\lambda + 6 = 36$$

$$\lambda = 12$$

$$\alpha = \frac{|\vec{c}||\vec{a}|}{|\vec{b}|} = \frac{3 \times 12}{2} \times \frac{12}{6}$$

$$\alpha = 36$$

Q7 (C)

By its given condition

: $\vec{a}, \vec{b}, \vec{c}$ are linearly independent vectors

$$\Rightarrow [\vec{a} \ \vec{b} \ \vec{c}] \neq 0 \quad \dots(i)$$

$$\text{Now, } [\vec{a} \ \vec{b} \ \vec{c}]$$

$$= \begin{vmatrix} 1+t & 1-t & 1 \\ 1-t & 1+t & 2 \\ t & -t & 1 \end{vmatrix}$$

$$C_2 \rightarrow C_1 + C_2$$

$$= \begin{vmatrix} 1+t & 2 & 1 \\ 1-t & 2 & 2 \\ t & 0 & 1 \end{vmatrix}$$

$$= 2 \begin{vmatrix} 1+t & 1 & 1 \\ 1-t & 1 & 2 \\ t & 0 & 1 \end{vmatrix}$$

$$= 2[(1+t) - (1-t) + t]$$

$$= 2[3t] = 6t$$

$$[\vec{a} \ \vec{b} \ \vec{c}] \neq 0 \Rightarrow t \neq 0$$

Q8 (B)

$$|\vec{a}| = 9 \text{ \& } (\vec{x}\vec{a} + \vec{y}\vec{b}) \cdot (6\vec{y}\vec{a} - 18\vec{x}\vec{b}) = 0$$

$$\Rightarrow 6xy|\vec{a}|^2 - 18x^2(\vec{a} \cdot \vec{b}) + 6y^2(\vec{a} \cdot \vec{b}) - 18xy|\vec{b}|^2 = 0$$

$$\Rightarrow 6xy(|\vec{a}|^2 - 3|\vec{b}|^2) + (\vec{a} \cdot \vec{b})(y^2 - 3x^2) = 0$$

This should hold $\forall x, y \in \mathbb{R} \times \mathbb{R}$

$$\therefore |\vec{a}|^2 = 3|\vec{b}|^2 \text{ \& } (\vec{a} \cdot \vec{b}) = 0$$

$$\text{Now } |\vec{a} \times \vec{b}|^2 = |\vec{a}|^2 |\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2$$

$$= |\vec{a}|^2 \cdot \frac{|\vec{a}|^2}{3}$$

$$\therefore |\vec{a} \times \vec{b}| = \frac{|\vec{a}|^2}{\sqrt{3}} = \frac{81}{\sqrt{3}} = 27\sqrt{3}$$

Q9 (B)

For angle to be acute

$$\vec{u} \cdot \vec{v} > 0$$

$$\Rightarrow a(\log_e b)^2 - 12 + 6a(\log_e b) > 0$$

$$\forall b > 1$$

$$\text{let } \log_e b = t \Rightarrow t > 0 \text{ as } b > 1$$

$$y = at^2 + 6at - 12 \text{ \& } y > 0, \forall t > 0$$

$$\Rightarrow a \in \phi$$

Q10 (A)

$$\vec{a} = 3\hat{i} + \hat{j}, \vec{b} = \hat{i} + 2\hat{j} + \hat{k}$$

$$\text{As } \vec{a} \times (\vec{b} \times \vec{c}) = \vec{b} + \lambda \vec{c}$$

$$\Rightarrow \vec{a} \cdot \vec{c} (\vec{b}) - (\vec{a} \cdot \vec{b}) \vec{c} = \vec{b} + \lambda \vec{c}$$

$$\Rightarrow \vec{a} \cdot \vec{c} = 1, \vec{a} \cdot \vec{b} = -\lambda$$

$$\Rightarrow (3\hat{i} + \hat{j}) \cdot (\hat{i} + 2\hat{j} + \hat{k}) = -\lambda$$

$$\Rightarrow \lambda = -5$$

Q11 (A)

$$\hat{a} \wedge \hat{b} = \frac{\pi}{4} = \phi$$

$$\hat{a} \cdot \hat{b} = |\hat{a}| |\hat{b}| \cos \phi$$

$$\hat{a} \cdot \hat{b} = \cos \phi = \frac{1}{\sqrt{2}}$$

$$\cos \theta = \frac{(\hat{a} + \hat{b}) \cdot (\hat{a} + 2\hat{b} + 2(\hat{a} \times \hat{b}))}{|\hat{a} + \hat{b}| |\hat{a} + 2\hat{b} + 2(\hat{a} \times \hat{b})|}$$

$$|\hat{a} + \hat{b}|^2 = (\hat{a} + \hat{b}) \cdot (\hat{a} + \hat{b})$$

$$|\hat{a} + \hat{b}|^2 = 2 + 2\hat{a} \cdot \hat{b}$$

$$= 2 + \sqrt{2}$$

$$\hat{a} \times \hat{b} = |\hat{a}| |\hat{b}| \sin \phi \hat{n}$$

$$\hat{a} \times \hat{b} = \frac{\hat{n}}{\sqrt{2}}$$

when $\hat{n}$ is vector $\perp$ $\hat{a}$ and $\hat{b}$

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let $\vec{c} = \vec{a} \times \vec{b}$ To practice more chapter-wise JEE Main PYQs, [click here to download the MARKS app](#) from Playstore

We know:

$$\vec{c} \cdot \vec{a} = 0$$

$$\vec{c} \cdot \vec{b} = 0$$

$$|\hat{a} + 2\hat{b} + 2\vec{c}|^2$$

$$= 1 + 4 + \frac{(4)}{2} + 4 \hat{a} \cdot \hat{b} + 8 \hat{b} \cdot \vec{c} + 4 \vec{c} \cdot \hat{a}$$

$$= 7 + \frac{4}{\sqrt{2}} = 7 + 2\sqrt{2}$$

Now

$$(\hat{a} + \hat{b}) \cdot (\hat{a} + 2\hat{b} + 2\vec{c})$$

$$= |\hat{a}|^2 + 2\hat{a} \cdot \hat{b} + 0 + \hat{b} \cdot \hat{a} + 2|\hat{b}|^2 + 0$$

$$= 1 + \frac{2}{\sqrt{2}} + \frac{1}{\sqrt{2}} + 2$$

$$= 3 + \frac{3}{\sqrt{2}}$$

$$\cos \theta = \frac{3 + \frac{3}{\sqrt{2}}}{\sqrt{2 + \sqrt{2}} \sqrt{7 + 2\sqrt{2}}}$$

$$\cos^2 \theta = \frac{9(\sqrt{2} + 1)^2}{2(2 + \sqrt{2})(7 + 2\sqrt{2})}$$

$$\cos^2 \theta = \left(\frac{9}{2\sqrt{2}} \right) \frac{(\sqrt{2} + 1)}{(7 + 2\sqrt{2})}$$

$$164 \cos^2 \theta = \frac{(82)(9)}{\sqrt{2}} \frac{(\sqrt{2} + 1)}{(7 + 2\sqrt{2})} \frac{(7 - 2\sqrt{2})}{(7 - 2\sqrt{2})}$$

$$= \frac{(82)(9)}{\sqrt{2}} [7\sqrt{2} - 4 + 7 - 2\sqrt{2}]$$

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Hints and Solutions

MathonGo

$$= 90 + 27\sqrt{2}$$

Q12 (D)

A(2, 3, 9) ; B(5, 2, 1) ; C(1, λ, 8) ; D(λ, 2, 3)

$$[\vec{AB} \ \vec{AC} \ \vec{AD}] = 0$$

$$\begin{vmatrix} 3 & -1 & -8 \\ -1 & \lambda - 3 & -1 \\ \lambda - 2 & -1 & -6 \end{vmatrix} = 0$$

$$\Rightarrow [-6(\lambda - 3) - 1] - 8(1 - (\lambda - 3)(\lambda - 2)) + (6 + (\lambda - 2)) = 0$$

$$3(-6\lambda + 17) - 8(-\lambda^2 + 5\lambda - 5) + (\lambda + 4) = 8$$

$$8\lambda^2 - 57\lambda + 95 = 0$$

$$\lambda_1 \lambda_2 = \frac{95}{8}$$

Q13 (C)

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$$|\vec{a}| |\vec{b}| |\vec{c}| = 14$$

$$\vec{a} \wedge \vec{b} = \vec{b} \wedge \vec{c} = \vec{c} \wedge \vec{a} = \theta = \frac{2\pi}{3}$$

$$\text{So, } \vec{a} \cdot \vec{b} = -\frac{1}{2}ab, \vec{b} \cdot \vec{c} = -\frac{1}{2}bc, \vec{a} \cdot \vec{c} = -\frac{1}{2}ac$$

(let)

$$(\vec{a} \times \vec{b}) \cdot (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{b})(\vec{b} \cdot \vec{c}) - (\vec{a} \cdot \vec{c})(\vec{b} \cdot \vec{b})$$

$$= \frac{1}{4}ab^2c + \frac{1}{2}ab^2c = \frac{3}{4}ab^2c$$

Similarly

$$(\vec{b} \times \vec{c}) \cdot (\vec{c} \times \vec{a}) = \frac{3}{4}abc^2$$

$$(\vec{c} \times \vec{a}) \cdot (\vec{a} \times \vec{b}) = \frac{3}{4}a^2bc$$

$$168 = \frac{3}{4}abc(a+b+c)$$

$$\text{So, } (a+b+c) = 16$$

Q14 (14)

$$|\vec{a} + \vec{b}|^2 = |\vec{a}|^2 + 2|\vec{b}|^2 ; \vec{a} \cdot \vec{b} = 3$$

$$\text{As } |\vec{a}|^2 + |\vec{b}|^2 + 2\vec{a} \cdot \vec{b} = |\vec{a}|^2 + 2|\vec{b}|^2$$

$$|\vec{b}|^2 = 2\vec{a} \cdot \vec{b} = 6$$

$$|\vec{a} \times \vec{b}|^2 = 75$$

$$|\vec{a}|^2 |\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2 = 75$$

$$6|\vec{a}|^2 - 9 = 75 \Rightarrow |\vec{a}|^2 = 14$$